

0-25 $\Rightarrow$ sub
 full marks in algo
 Join tel

Lec-25
 Master's
 cases

Lec_26

Practice Questions on Master's Theorem

↑ Right side is bigger
 case 1:

$$\frac{T(n)}{n^2} \mid \frac{n \log_b a}{n^3}$$

case 2:

$$n^2 \mid n$$

case 3:

$$n \mid n^p$$

case:

$$n \log^k n \mid n$$

$$T(n) = aT(n/b) + f(n)$$

$$T_c = \Theta(n^3)$$

$$T_c = \Theta(n^2)$$

$$T_c = \Theta(n \log n)$$

$$T_c = \Theta(n \log^k n)$$

• 1.) $T(n) = 16T(n/4) + n^{25}$

$T_C = \Theta(n^{25})$

$f(n) = n^{25}$

$a = 16$

$b = 4$

$n^{\log_b a}$

$T_C = \Theta(n^{25})$

$n^{\log_4 16}$

$n^{\log_4 16}$

$n^{\log_4 4^2}$

$2 \times \log_4 4$

$n^{2 \times 1}$

$n^{2 \times 1}$

n^2

15

$f(n)$
 n^{25}

n^2

n^{25}

$n^{\log_4 16}$

n^{25} / n^2

$\Theta(n^{25})$

$$b=2, a=2$$

$$n^{\log_b a} = n^{\log_2 2}$$

$$= n$$

$$\textcircled{2}$$

$$n^{\log_b a} = \underline{n}$$

- 2.) $T(n) = 2T(n/2) + n \cdot \log n^{25}$

①

$$f(n) = \underline{n(\log n)^{25}}$$

$$T_c = \textcircled{n \log^{26}}$$

3

• 3.) $T(n) = 2T(n/2) + n^2 \cdot \log n$

$\frac{n \log n}{n}$

$n^{\log_2 2} = n^{\log_2 2} = n^1 = n$

if you are getting answer
send me your JPL I will
pay so happy.

Case 2

Case 1

$f(n) = n^2 \log n$

$n^2 \log n$
 $n \times n \log n$

n

$TC = O(n^2 \log n)$

- 4.) $T(n) = 16T(n/4) + n^{25}$

$$f(n) = n^{25}$$

$$n \log_b a = n \log_4 16 = n^2$$

$$n \log_b a = n^2$$

$$= n^2$$

$$\Theta(n^{25})$$

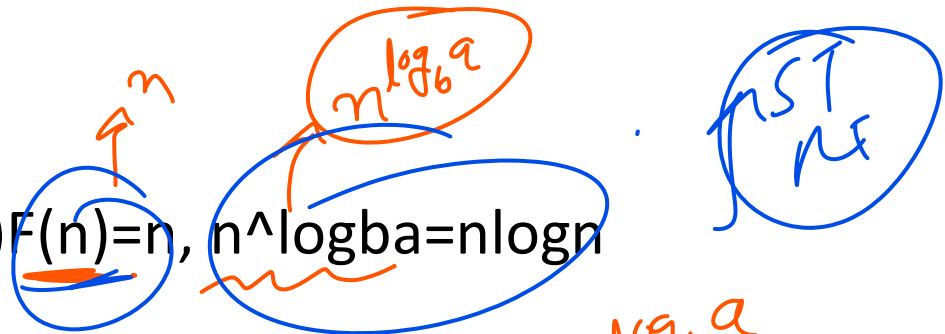
$$\Theta(n^2)$$

(ayer:2)





- 5.) $F(n)=n$, $n^{\log_b a} = n \log n$



$$\frac{f(n)}{n}$$

$$\frac{n \log_b a}{n \log n}$$

$$\frac{n \log n}{n \log n}$$

This is not any case

not n

case 1. 1, 2, 3, 4

$$\frac{n^2}{n^2} \mid \frac{n^3}{n}$$

$$\frac{n \log n}{n \log n} \mid n$$

$$\frac{n \log k}{n \log k} \mid \frac{n}{n}$$

can not be solved theorems

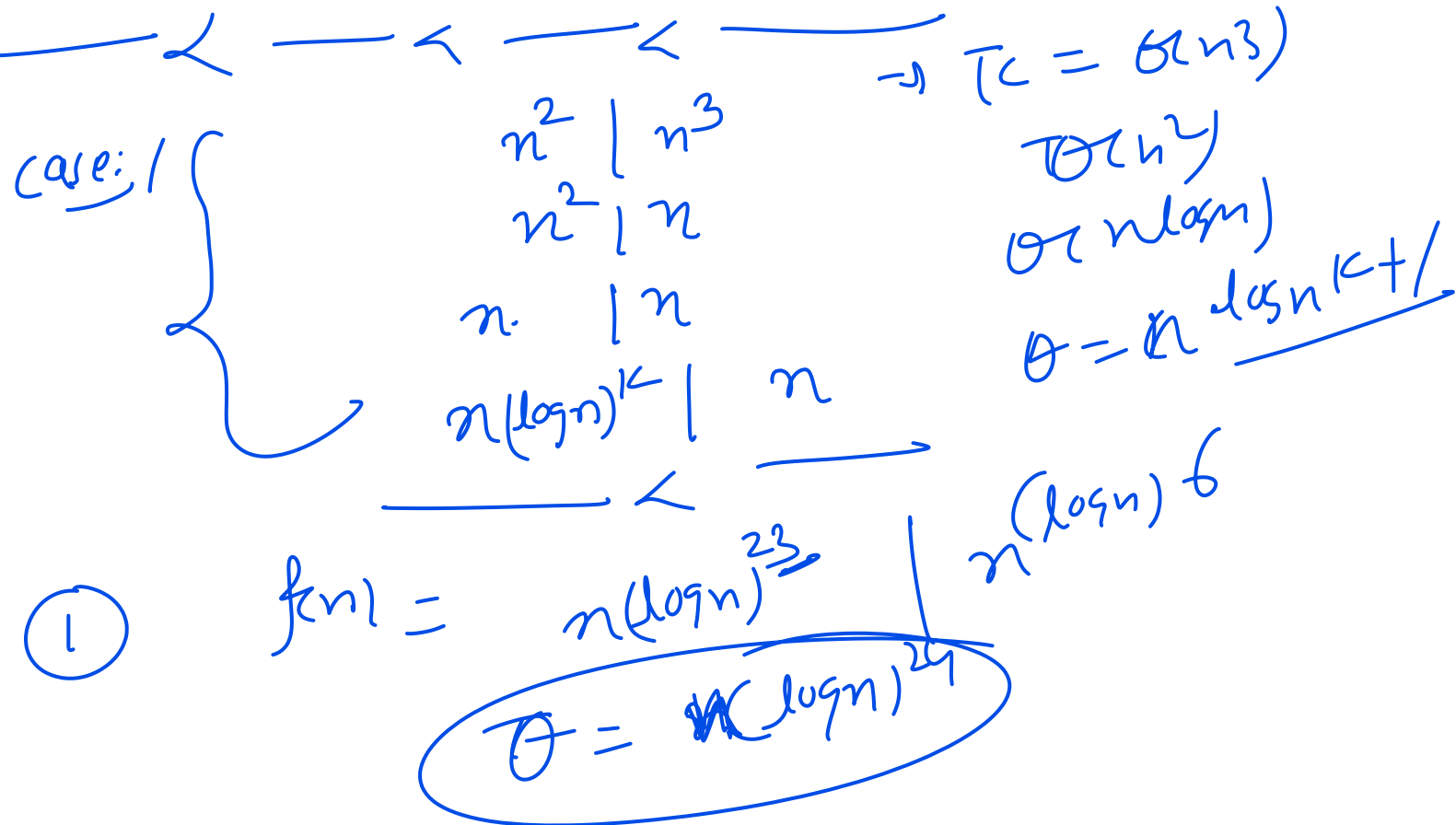
By the master

- 6.) $f(n)=n^2$, $n^{\log_b a} = n^3 \log n$

$$\begin{array}{r|l}
 f(n) & n^{\log_b a} \\
 n^2 & \\
 \hline
 n^2 & | \quad \frac{n^3 \log n}{n^2} = n \log n
 \end{array}$$

$$\begin{array}{l}
 TC = O(n^3 \log n) \\
 \text{Case 1:}
 \end{array}$$

Conclusion of Master theorem:



$$T(n) = T(n/2) + \log n \quad \text{Root}$$

Conclusion of Solving of recurrence Relation:

